

Secure networks with firewalls NAT, port filtering, packet filtering, and other methods

- 1) **Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.**

Step 1: Open Windows Defender Firewall

- Open Windows Security or search for Windows Defender Firewall with Advanced Security.
- Open the Advanced Security console and select Inbound Rules.

Step 2: Create the Inbound Rule

- Choose New Rule.
- Select Port as the rule type.
- Choose TCP and specify Specific local ports: 8080.
- Select Block the connection.
- Apply the rule to the required network profiles, such as Domain, Private, and Public, according to the assignment requirement.
- Give the rule a clear name such as Block TCP Port 8080 and finish the wizard.

Step 3: Verify

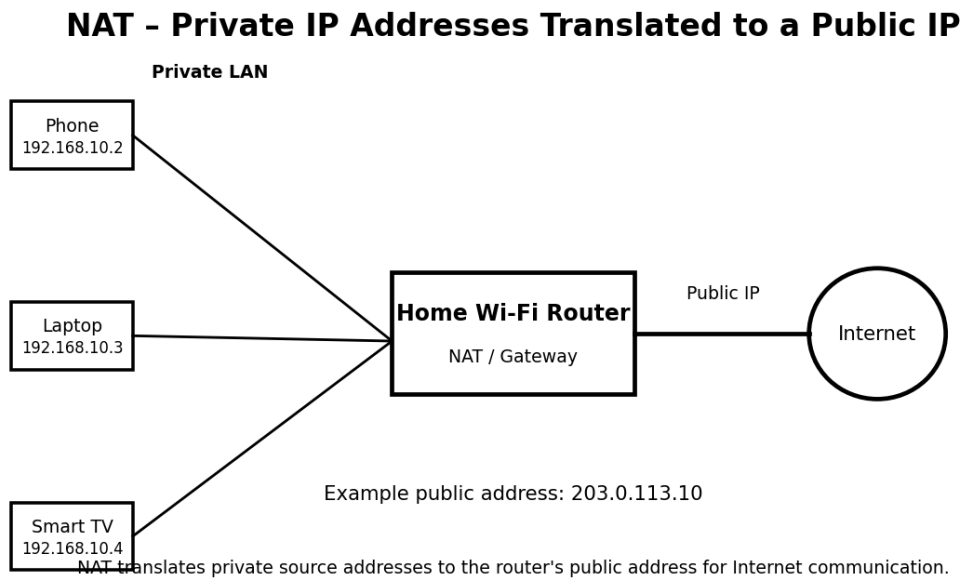
- Locate the newly created rule under Inbound Rules.
- Confirm that the rule is enabled and its Action is Block.

2) **Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for Internet access.**

NAT Explanation

- A home router normally uses a private LAN address range for devices inside the home network.
- When a device accesses the Internet, the router performs NAT so traffic can use the router's public Internet address.
- Example private addresses: Phone 192.168.10.2, Laptop 192.168.10.3, Smart TV 192.168.10.4.
- Example public address used in the diagram: 203.0.113.10.

NAT Diagram



- The reference diagram below shows the three private devices, the router performing NAT, and the public Internet side.

- 3) Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.**

Step 1: Open Router Administration

- Connect to the router and open its administration interface.
- Locate Firewall, Access Control, Security, Port Filtering, or a similarly named section.

Step 2: Create the Filtering Rule

- Create an outbound/egress rule.
- Select TCP traffic and specify destination port 25 (SMTP).
- Set the action to Block/Deny.
- Apply the rule to the required LAN devices or the whole LAN, depending on the router's features.
- Save and enable the rule.

Why Block Port 25?

- Blocking outbound TCP port 25 can reduce the ability of compromised devices to send email directly to external mail servers.
- It can therefore help limit some forms of spam or malware-driven direct SMTP abuse, while legitimate mail services may use authenticated submission ports such as 587.

- 4) Given a sample packet filtering rule set, identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53.

Important Note

- The screenshot does not provide the trainer's actual sample packet-filtering rule set. Therefore, the exact allow/block result for the three packets cannot be determined from the supplied task alone.
- Use the source/destination IP addresses and destination ports together with the trainer's rule set to determine the result.

Packet Analysis Method

- Packet 1 — HTTP: Source 192.168.1.5 → Destination 172.217.166.78:80. Check whether the rules allow or deny traffic to destination port 80 from that source.
- Packet 2 — SSH: Source 10.0.0.10 → Destination 34.201.23.1:22. Check whether the rules allow or deny destination port 22 from that source.
- Packet 3 — DNS: Source 192.168.1.7 → Destination 8.8.8.8:53. Check whether the rules allow or deny DNS traffic to port 53 from that source.

Result Table

- Packet 1 — HTTP — Result: _____ — Rule matched: _____.
- Packet 2 — SSH — Result: _____ — Rule matched: _____.
- Packet 3 — DNS — Result: _____ — Rule matched: _____.

5) Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.

Method 1: Restrict Unnecessary Ports and Services

- Firewalls can allow only required network services and block unnecessary inbound ports.
- This reduces the number of externally reachable services and limits opportunities for unauthorized access.

Method 2: Filter Traffic by IP, Port, and Protocol

- Network security controls can restrict traffic based on source/destination IP addresses, ports, and protocols.
- Rules can help block suspicious or unauthorized traffic while allowing legitimate application traffic.

Method 3: Limit and Detect Abusive Traffic

- Firewalls and related network controls can help enforce traffic policies and work with other security systems to identify or limit suspicious patterns.
- For large Internet services, these controls can be combined with load balancers, DDoS protection, intrusion detection/prevention, and application-layer security.

Summary

- Large online services such as Instagram or Zomato use layered security rather than relying on a single firewall rule.
- Firewall and packet-filtering controls are one part of a broader defense strategy that helps protect servers and services from unauthorized or abusive traffic.